The Concept of Creating a Software Digital Twin of an Electronic Device

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***Abstract*—The concept of creating a software digital twin of electronic devices is considered. This digital twin can be used as the basis of virtual simulators that are part of electronic teaching and training aids in the field of operation of high-tech products (electronic devices). Definitions, goals, problems of the electronic device digital twin are given. Also, digital twin structure and a variant of its implementation using web technologies are introduced.**

Keywords—digital twin, web simulator, e-tutorial, web technology, electronic devices

I. Introduction

The requirements for the level of training and education of specialists in the nuclear weapon complex of the Russian Federation, in particular, specialists in the field of improving the operation and maintenance of high-tech products (electronic devices and radio-electronic equipment (REE)) have risen due to a rapidly changing global political situation and due to an increase in the number and complexity of electronic devices being developed. At the same time, software and information support for the process of preparing these specialists should answer the following purpose: being cross-platform, support for import substitution; high speed of development and implementation of software and information products, scalability.

II. Electronic Device Digital Twin

The authors create electronic teaching and training aids (ETA) designed for theoretical and practical training of specialists and maintenance personnel [1,2,3] in accordance with the concept of developing new methods for training specialists in the nuclear industry as well as improving existing ones and adopting new promising automation and technical training methods taking into account a systematic approach. Also, authors create ETA to introduce specialists with training processes in the field of electronic devices operation based on design and operational documentation (DD and OD) for the products created by the department (electronic devices and REE).

ETA are developed by the authors based on web technologies which allows to create modern interactive software and information products with a high degree of flexibility and scalability quickly. The ETA main part developed by the authors is virtual (software or web) simulators. These simulators are software and information products designed for practical training specialists and simulation of operation processes in order to develop skills and abilities when working with complex, high-tech and expensive products (electronic devices) [5,6].

Virtual computer simulators of electronic devices demonstrate the these devices operation modes in a special interactive mode allowing training specialists in the field of products operation in connection with the simulating possibility a wide range of problems that arise during the operation of devices and electronic equipment [4,6].

The key elements of virtual software and web simulators that are ETA part are electronic devices digital twins. The concept of them is proposed and implemented by the present report authors.

We start with considering general information on products digital twins.

A digital twin (hereinafter referred to as DT) is a system consisting of a product digital model and two-way data connection with the product and its components [8]. A product digital model is a system of mathematical and computer models, as well as product electronic documents that describe the structure, functionality, and behavior of a newly developed or operated product at various stages of its life cycle [8].

The DT aims include the following:

- An assessment of the scientific validity, feasibility and technological feasibility of the developed or finished product in various operating conditions;
- An ability to trace the adopted technical solutions at various stages of the product life cycle;
- Reducing the cost and terms of DD and OD development, prototypes and testing of the product and its components [8].

DT combines a mathematical and computer model of a product (in this case, an electronic device). It means that DT allows to describe the electronic device as well as it has been specified by the customer and to approach the development, production and operation of complex, high-tech and expensive products within the framework of a systematic approach [2]

One can understand a certain entity under the electronic device DT (hereinafter we will refer to it as EDDT). It is a

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virtual electronic device that has the same visual and algorithmic characteristics as a real device. The EDDT looks like a real device (in computer design) and all operation modes of this device should work properly at the same time.

The authors propose the structure of the EDDT within the framework of the concept of DT creating (Fig. 1).

Fig. 1. The structure of the electronic device digital twin

The EDDT consists of two blocks. They are the block of an electronic device functional-algorithmic model (FAM) and the block of an electronic device virtual interface (VI).

The FAM of EDDT contains programmed algorithms for the considered virtual electronic device operation. The FAM is the central element ("core") of the DT. Also, the FAM accumulates the data about all device operation modes (as well as about all possible options for this device interaction with the user (operation specialist, maintenance personnel) or with other devices according to the OD) in the form of software-information algorithms [5].

The electronic device software VI block (the EDVI block) is intended for the implementation of virtual (with the help of a computer) user interaction with this virtual electronic device. EDVI should looks like a real device (in computer design) with the all structural elements that are significant for the user (operator) (buttons, indicators, LEDs). Moreover, all interactive zones should be highlighted.

The user affects the interactive zones of the virtual electronic device VI in accordance with the OD for this device. Then the DT VI processes and displays the user's action on the computer screen, producing a computer simulation of this action (a computer animation-graphic simulation of pressing the buttons on the device front panel, turning on/off the toggle switches or other actions).

At the same time, the data about the user's action is transmitted from the DT VI block to the FAM block where this user action (including the erroneous one) is processed in accordance with the programmed algorithms for the product operation. After that an answerback from the FAM (updated attributes, properties, and states of the FAM) is shown at the DT VI.

Then the VI program codes display this answerback of the corresponding virtual electronic device producing a computer animation-graphic simulation. For example, it can be the text appearance on the virtual electronic device display, turning on/off indicator LEDs on the device front panel, etc. [5].

It is obvious that a single virtual simulator may include the DT of several electronic devices interacting with each other.

The result of the FAM and DT VI blocks separation provides the possibility of scaling and parallel development of these blocks. The generated and programmed FAM and VI of virtual electronic devices can be used in various combinations repeatedly and quickly in the development of both web simulators and software simulators at the request of the customer. It reduces the time and laboriousness of developing both virtual simulators and the ETA as a whole significantly.

We can conclude as a result of the functionality study of the DT (which is a virtual electronic device in computer form) that a EDDT can be used not only for training and teaching personnel to work with real electronic devices (such digital devices can also be used on a development work stage for testing DD and OD for devices), but also as an auxiliary tool for developing a friendly user interface for REE products. Indeed, the development of a CD of an electronic device is several orders of magnitude lower than the cost of manufacturing real prototypes of REA products, and the study and analysis of the functioning of a CD of an electronic device, its FAM and VI, bypassing the process of manufacturing expensive prototypes and prototypes, can be useful for developers of real electronic devices in the process of developing design documentation and ED at the stages of development and research work.

We consider the variant of the software implementation of the electronic device CD proposed by the authors.

The avalanche acceleration of scientific and technological progress in the field of information and software technologies has led to a situation where this area complex and time-consuming solutions are rapidly becoming obsolete, not having time to be implemented and introduced into production processes. It is obvious that the significance and importance of high-tech, science-intensive, time-consuming and costly development of solutions (in the field of virtual and augmented reality, supercomputer modeling) is not in doubt. However, it is necessary to choose a technology that allows you to create easy-to-design, easily scalable and easily modified cross-platform software and information products that can be quickly adjusted/finalized in accordance with changing customer requirements in order to increase adaptability to the rapidly changing environment of domestic software and information development,.

The web technologies use can help in solving such problems. Their advantages are cross-platform (ensuring functioning on different operating systems like Microsoft Windows, Unix, etc.), scalability, the ability to work both on stand-alone computers and in the enterprise or the Internet computer network, application development speed, no requirements for the presence of a programming environment, launching applications using a standard browser on the user's computer, etc.

In accordance with the foregoing, the EDDT components (FAM and VI) can certainly be created both as software applications (for example, in the C++ programming language using the Qt framework) and using software web technologies (such as JavaServerPages, PHP). The experience of ETA developing based on the DT has shown that the DT VI (as well

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as a simplified FAM version) can also be developed using the HTML/CSS web technology with or without the use of Javascript [7].

The generalized ETA structure developed by the authors is shown in Fig. 2. This structure takes the proposed concept of EDDT creating into account.

Fig. 2. The structure of the educational training and teaching aids with a central element (the electronic device digital twin)

The authors created the electronic device system ETA (developed by the department) using the EDDT creating proposed concept. This ETA contains a web simulator, the central part of which was the EDDT prototypes. The developed ETA was approved for serial production [5,6] after its acceptance by the customer and passing interdepartmental tests with the assignment of the letter "O1". Similar ETA developed using web technologies and with virtual web simulators included in them (which are based on the considered EDDT) can be used both for theoretical training of specialists (as interactive illustrations of lectures) and for practical training which means the formation of stable and practical skills in the operating devices process, devices preparation and operating devices modes [9-14].

The EDDT introduction in the process of the electronic device life cycle allows increasing the electronic device and REE competitiveness at the stage of development and research work by improving the quality and reducing development time in addition to improving the quality and efficiency of education and training of specialists at the operational stage. It is possible by improving quality and reducing development time of the DD and OD. This provides the opportunity to study, simulate and analyze the electronic device functioning in real time up to the stage of manufacturing expensive prototypes, significantly reducing the cost of product development.

REFERENCES

[1] Vedernikov, V.L., Gorbatenko, N.V., Martynov, A.P. Electronic training aids at the stages of a life cycle of devices and radio-electronic equipment // Proceedings of XXXVIII All-Russian Science and Technology Conference “Problems of efficiency and safety of functioning of complex technical and information systems”.– Serpukhov. – 2019.

[2] Phomchenko, V.N., Vedernikov, V.L., Volkov, K.O., Gorbatenko, N.V. Interactive Web Training Simulator. Certificate of State registration of discovery № 2723365, June 10, 2020. https://elibrary.ru/item.asp?id=43902191.

[3] Phomchenko, V.N., Vedernikov, V.L., Martynov, A.P., Gorbatenko, N.V., Splyukhin, D.V.,Osin, D.V., Dun’kovich V.V. The program of procedural web simulator of a typical device of radio-electronic equipment, version 1.0. Certificate of State registration of the computer program № 2019666228, December 6, 2019. https://elibrary.ru/item.asp?id=41532933.

[4] All-Union State Standard (GOST) P 57700.37-2021. Computer models and modeling. Digital twins of products. General provisions.

[5] Martynova, I.A., Martynov, A.P., Nikolaev, D.B. Cryptographic systems and method of factorial information reduction // Izvestiya Instituta inzhenernoy phiziki. 2016. №4(42). P. 54–57. https://elibrary.ru/item.asp?id=27697857.

[6] Martynova, I.A., Martynov, A.P.. Permutation functions in the number system of a number of factorial sets // Proceedings of Voronezh State University. Series: System analysis and information technologies. 2016. № 3. P. 42–49. https://elibrary.ru/item.asp?id=27296781

[7] Martynova, I.A., Splyukhin D.V. Investigation of criterion factors in the expansion of series of factorial sets // Nauka. Mysl’. 2017. № 5. P. 13-16.

[8] Modern methods of ensuring safety in the nuclear power industry: Monograph / Gribunin, V.G., Kostyukov, V.E., Martynov, A.P., Nikolaev, D.B., Phomchenko, V.N. // Edited by Astajkin, A.I. Sarov: FSUE “RFNC-ASRIEP” (FGUP “RFYaTs-VNIIEF”), 2014. 636 pages. https://elibrary.ru/item.asp?id=41589697.

[9] Martynov, A.P., Nikolaev, D.B., Splyukhin, D.V., Phomchenko, V.N., Martynova, I.A. Software package for the analysis of substitutions for a number of factorial sets. Certificate of State registration of the computer program № 2020613795. Date of registration March 23, 2020. Application № 2020612500, March 3, 2020. https://elibrary.ru/item.asp?id=42710013.

[10] Patent RF 2623894, 07.09.2017. Data transformation method with equiprobability initialization / Martynov, A.P., Martynova, I.A., Marunin, M.V., Nikolaev, D.B., Phomchenko, V.N. Certificate of State registration of discovery № RU 2623894 C1, publication date June 29, 2017. https://elibrary.ru/item.asp?id=38266740.

[11] Martynova, I.A., Martynov, A.P., Nikolaev, D.B., Phomchenko, V.N. System of dimension-group transformation of information streams (Sistemy prostranstvenno-gruppovogo preobrazovaniya informatsionnykh potokov) // Problems of Atomic Science and Technology (VANT), Series: cep. Mathematical modeling of physical processes. 2022. Issue 1. P.70-82.

[12] Volkov, K.O., Martynov, A.P., Martynova, I.A., Nikolaev D.B., Nikolaeva, I.A., Phomchenko, V.N. A method for dynamically transforming data during storage and transmission. Certificate of State registration of discovery № RU 2699589 C1, publication date September 6, 2019. Application № 2019103419, date of registration February 7, 2019. https://elibrary.ru/item.asp?id=39535506.

[13] Volkov, K.O., Martynov, A.P., Martynova, I.A., Nikolaev D.B., Nikolaeva, I.A., Phomchenko, V.N. Method for generating identification features of a group of objects. Certificate of State registration of discovery № RU 2700401 C1, publication date September 16, 2019. Application № 2019107851, date of registration March 19, 2019. https://elibrary.ru/item.asp?id=40881696.

[14] Martynova, I.A.. The theoretic-functional analysis of transformation functions of information cryptographic systems // Izvestiya Instituta inzhenernoy phiziki. – 2020. – № 4 (58). – P. 73–77. https://elibrary.ru/item.asp?id=44483935.

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